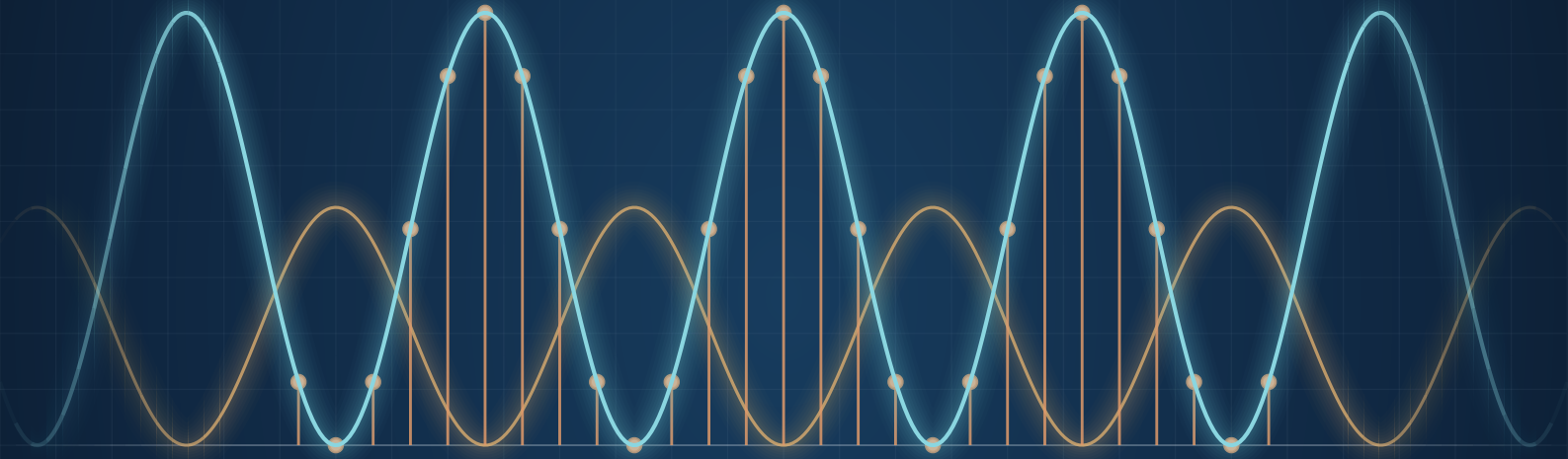




QUANTUM COMPUTING

# Quantum Computing

*Student Workbook*



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## QUANTUM COMPUTING

# Student Workbook

*Every question in the course, with no answer and no solution. Work each one on the page, then check it against the artifact or against the instructor edition.*

| Contains                       | Level         | Answers                     |
|--------------------------------|---------------|-----------------------------|
| 120 questions across 6 modules | Undergraduate | Not printed in this edition |

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This course is an adaptation. Its syllabus and the sequence of topics it teaches derive from **Quantum Computing Lectures** by Aleksandr Krasnok, [github.com/AlexKrasnok/quantum-computing-lectures](https://github.com/AlexKrasnok/quantum-computing-lectures), used under [CC BY 4.0](#). Every scene, figure, laboratory and question here is written for this artifact.

**How to use it**

The questions are in the order the course meets them, and each is labelled with what it asks for. Only the statement and its lettered parts are printed; the reasoning stays for you to supply. The question numbers are shared with every other edition, so D5-04 is the same question in the artifact, in this workbook and in the instructor solutions.

## MODULE

## 1

# The Mathematics of Quantum States

20 questions on the mathematics of quantum states. Write your reasoning in the space under each one.

## D1-01 · INNER PRODUCTS, NORMS AND ORTHOGONALITY

Two states of one qubit are

$$|a\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix}, \quad |b\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

- Compute  $\langle a|b\rangle$ .
- Compute  $\langle b|a\rangle$ , and state the relation between the two.
- Give  $|\langle a|b\rangle|^2$  and say what it means about telling the two states apart.

## D1-02 · INNER PRODUCTS, NORMS AND ORTHOGONALITY

An unnormalised column is  $|\tilde{\psi}\rangle = 3|0\rangle + 4i|1\rangle$ .

- Give  $\langle \tilde{\psi}|\tilde{\psi}\rangle$  and the length  $\|\tilde{\psi}\|$ .
- Give the normalised state  $|\psi\rangle$ .
- Give the two computational-basis probabilities that Chapter 2 will read off it.

## D1-03 · INNER PRODUCTS, NORMS AND ORTHOGONALITY

A student computes the squared length of  $|a\rangle = (1, i)$  in NumPy with `np.dot(a, a)` and gets zero.

- Compute  $\langle a|a\rangle$  correctly.
- Compute the quantity `np.dot(a, a)` returns, and say what it would claim about  $|a\rangle$ .
- Give the normalised state.

## D1-04 · EXPANDING A STATE IN A BASIS

A state is  $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$ .

- a) Confirm that it is normalised.
- b) Expand it in the basis  $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$ .
- c) Give  $|\langle +|\psi\rangle|^2$  and  $|\langle -|\psi\rangle|^2$ , and check them.

## D1-05 · EXPANDING A STATE IN A BASIS

The resolution of the identity in the  $X$  basis is  $|+\rangle\langle +| + |-\rangle\langle -| = I$ .

- a) Insert it into  $\langle 0|1\rangle$  and write the resulting sum.
- b) Evaluate every factor and give the total.
- c) Say what the calculation demonstrates.
- d) For a complete function basis with coefficients  $c_n$ , state Parseval identity and the squared error after retaining only  $n \leq N$ .

## D1-06 · GLOBAL AGAINST RELATIVE PHASE

Three states are

$$|\psi_1\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \quad |\psi_2\rangle = \frac{i|0\rangle - |1\rangle}{\sqrt{2}}, \quad |\psi_3\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}.$$

- a) Decide whether  $|\psi_1\rangle$  and  $|\psi_2\rangle$  are the same physical state.
- b) Compute  $\langle \psi_1 | \psi_3 \rangle$ .
- c) Say which pair a measurement can separate with certainty, and which it cannot separate at all.

## D1-07 · GLOBAL AGAINST RELATIVE PHASE

A qubit is prepared in  $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\varphi}|1\rangle)$ .

- Give  $P(0)$  and  $P(1)$  in the computational basis.
- Apply a Hadamard and give the two probabilities after it, as a function of  $\varphi$ .
- Evaluate both at  $\varphi = \pi/3$ .

## D1-08 · BUILDING AN OPERATOR AND TESTING IT

A state is  $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ .

- Form the operator  $P = |\psi\rangle\langle\psi|$  as a  $2 \times 2$  matrix.
- Verify that  $P^\dagger = P$  and  $P^2 = P$ .
- Compute  $P|0\rangle$  and give its length.

## D1-09 · BUILDING AN OPERATOR AND TESTING IT

Two matrices are

$$A = \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix}, \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

- Decide whether each is Hermitian.
- Compute  $AX$  and  $XA$ .
- Decide whether  $AX$  is Hermitian, and say what settles it.

## D1-10 · BUILDING AN OPERATOR AND TESTING IT

A matrix is  $U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}$ .

- Write  $U^\dagger$ .
- Compute  $U^\dagger U$  and decide whether  $U$  is unitary.
- Apply  $U$  to  $|0\rangle$  and confirm that the length is preserved.

## D1-11 · BUILDING AN OPERATOR AND TESTING IT

A matrix is  $M = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ .

- Give  $\det M$ .
- Compute  $M^\dagger M$ .
- Apply  $M$  to  $|1\rangle$  and give the length of the result. Is  $M$  a legal gate?

## D1-12 · EIGENVALUES, THE SPECTRAL FORM, AND FUNCTIONS

A Hermitian matrix is  $A = \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}$ .

- Give the characteristic equation and solve it.
- Find a normalised eigenvector for each eigenvalue.
- Check the two eigenvalues against the trace and the determinant.

## D1-13 · EIGENVALUES, THE SPECTRAL FORM, AND FUNCTIONS

Use the eigenvectors of  $A = \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}$ , namely  $|v_+\rangle = \frac{1}{\sqrt{2}}(1, i)$  with  $\lambda = 2$  and  $|v_-\rangle = \frac{1}{\sqrt{2}}(1, -i)$  with  $\lambda = 0$ .

- Build the projectors  $P_+$  and  $P_-$ .
- Verify  $P_+ + P_- = I$  and  $P_+ P_- = 0$ .
- Reassemble  $A$  from the spectral form and confirm it.

## D1-14 · EIGENVALUES, THE SPECTRAL FORM, AND FUNCTIONS

Continue with  $A = \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} = 2P_+ + 0P_-$ .

- a) Give  $A^2$  from the spectral form, and confirm it by squaring the matrix.
- b) Give  $\sqrt{A}$ , meaning the positive operator whose square is  $A$ .
- c) Give  $e^{-iAt}$ .

## D1-15 · EIGENVALUES, THE SPECTRAL FORM, AND FUNCTIONS

For  $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ , compute  $e^X$ .

- a) Give the eigenvalues and the two projectors of  $X$ .
- b) Give  $e^X$  from the spectral form, in closed form and to four decimals.
- c) Give the matrix that exponentiating  $X$  entry by entry would produce, and show by its eigenvalues that it cannot be the exponential of any Hermitian matrix.

## D1-16 · TENSOR PRODUCTS, BIT ORDER AND COUNTING

A two-qubit register has  $q_1$  in  $|1\rangle$  and  $q_0$  in  $|0\rangle$ .

- a) Write the joint state as a four-entry column, in the ordering this course fixes.
- b) Say which entry carries the amplitude, and why.
- c) Give the column the opposite convention would produce, and name the state it describes.

## D1-17 · TENSOR PRODUCTS, BIT ORDER AND COUNTING

Consider

$$|\alpha\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle-|1\rangle}{\sqrt{2}}, \quad |\beta\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}.$$

- Write  $|\alpha\rangle$  as a four-entry column.
- Show that  $|\beta\rangle$  cannot be written as a product of two single-qubit states.
- Arrange the four amplitudes of each state as a  $2 \times 2$  array  $c_{q_1 q_0}$  and give its determinant.

## D1-18 · TENSOR PRODUCTS, BIT ORDER AND COUNTING

A register holds  $n$  qubits, and a simulator stores each complex amplitude as two double-precision numbers.

- Give the number of amplitudes for  $n = 1, 2, 3, 4$ .
- Give the number of amplitudes and the storage in bytes for  $n = 20$ .
- Give the number of bits one measurement of the register returns, and say what that implies for an algorithm.

## D1-19 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES

A qubit is prepared in

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle, \quad \theta = \frac{\pi}{3}, \quad \varphi = \frac{\pi}{2}.$$

- Give the state as an explicit column.
- Give  $P(0)$  and  $P(1)$ .
- Give  $P(+)$  and  $P(-)$ , where  $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$ .
- The state is now multiplied by  $e^{i\pi/4}$ . State which of the four probabilities change.

## D1-20 · A FULL-LENGTH QUESTION COMBINING SEVERAL OF THE SHAPES

An operator is

$$G = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

- a) Decide whether  $G$  is Hermitian.
- b) Compute  $G^2$ , and use it to give the eigenvalues of  $G$  without solving the characteristic equation.
- c) Give  $e^{-iGt}$  in closed form.
- d) Evaluate it at  $t = \pi/(2\sqrt{2})$  and confirm that the result is unitary.

## MODULE

## 2

# States, Measurement and Dynamics

20 questions on states, measurement and dynamics. Write your reasoning in the space under each one.

## D2-01 · BORN PROBABILITIES IN A NAMED BASIS

A qubit is prepared in  $|\psi\rangle = \frac{1}{5} (3|0\rangle + 4i|1\rangle)$ .

- Give the probabilities of the two outcomes of a  $Z$  measurement.
- Give the probabilities of the two outcomes of an  $X$  measurement.
- Is the state an eigenstate of either? Say how you can tell without solving anything.

## D2-02 · BORN PROBABILITIES IN A NAMED BASIS

A qubit in  $|0\rangle$  is measured along the direction  $\mathbf{n} = (\sin \alpha, 0, \cos \alpha)$ .

- Write the two projectors of that measurement.
- Give  $p(+1)$  as a function of  $\alpha$ , and evaluate it at  $\alpha = 60^\circ$ .
- At which  $\alpha$  is the measurement a fair coin?

## D2-03 · BORN PROBABILITIES IN A NAMED BASIS

A qubit is prepared in  $|+i\rangle = \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)$ .

- Give the outcome probabilities for a  $Z$  measurement.
- Give them for an  $X$  measurement.
- Give them for a  $Y$  measurement, and say what the three answers together mean.

## D2-04 · THE STATE AFTER A MEASUREMENT, AND SEQUENCES

A qubit is prepared in  $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$  and measured in the  $X$  basis.

- a) Give the two outcome probabilities.
- b) Give the state after each outcome.
- c) The outcome was  $+1$ . A  $Z$  measurement now follows. Give its probabilities.

## D2-05 · THE STATE AFTER A MEASUREMENT, AND SEQUENCES

A qubit is prepared in  $|0\rangle$ . Three measurements are made in a row:  $Z$ , then  $X$ , then  $Z$  again.

- a) Give the outcome of the first measurement and its probability.
- b) Give the probabilities of the second, and the state after each.
- c) Give the probability that the third measurement agrees with the first.
- d) Give the same probability when the middle measurement is omitted.

## D2-06 · THE STATE AFTER A MEASUREMENT, AND SEQUENCES

A readout reports the wrong bit with probability  $\epsilon = 0.05$ , in either direction, so its effects are  $E_0 = (1 - \epsilon)|0\rangle\langle 0| + \epsilon|1\rangle\langle 1|$  and  $E_1 = I - E_0$ .

- a) Give the probability of reporting 0 for a qubit truly in  $|0\rangle$ .
- b) For a state with true  $p(0) = q$ , give the reported probability.
- c) A long run reports 0 a fraction 0.62 of the time. Give the corrected estimate of  $q$ , and say what the correction does to the error bar.
- d) Do the effects determine the state after outcome  $m$ ? Name the extra object and give its update rule.

## D2-07 · EXPECTATION VALUES AND VARIANCES

A qubit is in  $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$ .

- Give  $\langle Z \rangle$ .
- Give  $\langle X \rangle$  and  $\langle Y \rangle$ .
- Give the Bloch vector and check it.

## D2-08 · EXPECTATION VALUES AND VARIANCES

Continue with  $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$ , for which  $\langle Z \rangle = \frac{1}{2}$ ,  $\langle X \rangle = \frac{\sqrt{3}}{2}$  and  $\langle Y \rangle = 0$ .

- Give  $\text{Var}(Z)$  and  $\Delta Z$ .
- Give  $\text{Var}(X)$  and  $\Delta X$ .
- Give  $\Delta X \Delta Z$  and the Robertson bound, and say what the comparison shows.

## D2-09 · EXPECTATION VALUES AND VARIANCES

An observable and a state are

$$H = \begin{bmatrix} 2 & 1-i \\ 1+i & 3 \end{bmatrix}, \quad |\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix}.$$

- Give  $\langle H \rangle$ .
- Give  $\langle H^2 \rangle$  and  $\text{Var}(H)$ .
- Give the eigenvalues of  $H$ , and say what they show about the answer to (a).

## D2-10 · COMMUTATORS, COMPATIBILITY AND THE BOUND

Use the Pauli product rule  $\sigma_i \sigma_j = \delta_{ij} I + i \varepsilon_{ijk} \sigma_k$ .

- Give  $[X, Y]$ ,  $[Y, Z]$  and  $[Z, X]$ .
- Give the anticommutator  $\{X, Y\}$ .
- Do  $X$  and  $Y$  share an eigenvector? Say how the answer follows from (a).

## D2-11 · COMMUTATORS, COMPATIBILITY AND THE BOUND

Two pairs of observables are  $A = Z$  with  $B = 3Z + 2I$ , and  $A = Z$  with  $B = X$ .

- a) Give  $[Z, 3Z + 2I]$ .
- b) Give the common eigenvectors of the first pair, with the eigenvalue of each observable.
- c) Give  $[Z, X]$ , and say whether any state makes both readings certain.

## D2-12 · COMMUTATORS, COMPATIBILITY AND THE BOUND

A qubit is in  $|+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$ .

- a) Give  $\langle X \rangle$ ,  $\langle Y \rangle$  and  $\langle Z \rangle$ .
- b) Give  $\Delta X$  and  $\Delta Z$ .
- c) Compare  $\Delta X \Delta Z$  with  $\frac{1}{2} |\langle [X, Z] \rangle|$ .

## D2-13 · EVOLUTION, STATIONARY STATES AND PULSES

A qubit evolves under  $H = \frac{\omega}{2} Z$  from the initial state  $|+\rangle$ .

- a) Give  $U(t)$ .
- b) Give  $|\psi(t)\rangle$ , with the global phase removed.
- c) Give  $P(0)$  and  $P(+)$  as functions of  $t$ .
- d) Give the first time at which the state is  $|-\rangle$ .

## D2-14 · EVOLUTION, STATIONARY STATES AND PULSES

A system has  $H|E_1\rangle = E_1|E_1\rangle$  and  $H|E_2\rangle = E_2|E_2\rangle$ , with  $E_2 > E_1$ .

- Show that  $|E_1\rangle$  is stationary.
- Give  $|\psi(t)\rangle$  for the initial state  $\frac{1}{\sqrt{2}}(|E_1\rangle + |E_2\rangle)$ , with the global phase removed.
- Give the period of any observable quantity, and say what happens to it when both energies are raised by the same amount.
- For an infinite square well of width  $a$ , give  $\phi_n(x)$  and  $E_n$ , and show that  $E_2 = 4E_1$ .
- In the coordinate representation, give  $\hat{p}$  and the free-particle Hamiltonian. Why is one plane wave not a normalisable physical state on the whole line?

## D2-15 · EVOLUTION, STATIONARY STATES AND PULSES

A qubit starting in  $|0\rangle$  is driven on resonance by  $H = \frac{\Omega}{2}X$ .

- Give  $U(t)$ .
- Give  $P(1)$  as a function of  $t$ .
- Give the first time at which the qubit has been flipped, and the operator  $U$  is then equal to.
- Give the first time at which the state is an even superposition.

## D2-16 · EVOLUTION, STATIONARY STATES AND PULSES

The same qubit is driven off resonance, by  $H = \frac{1}{2}(\Omega_x X + \Delta Z)$  with  $\Omega_x = 1$  and  $\Delta = 1$ , starting from  $|0\rangle$ .

- Give the generalised rate  $\Omega$  and the rotation axis  $\mathbf{n}$ .
- Give the largest value  $P(1)$  ever reaches.
- Give the first time at which it reaches that value.
- Say what a  $\pi$  pulse calibrated on resonance does here.

## D2-17 · WHAT A FINITE RUN IS WORTH

A circuit whose true outcome probability is near  $\frac{1}{2}$  is run for  $N = 10\,000$  shots.

- Give the standard error of the estimated probability.
- Give the number of shots needed to reach a standard error of 0.001.
- The readout has a known bias of 0.02. Say what running more shots does to it.

## D2-18 · WHAT A FINITE RUN IS WORTH

The expectation value of a Pauli observable is estimated from  $N$  shots by averaging the  $\pm 1$  readings. The true value is  $\langle X \rangle = 0.8$ .

- Give  $p(+1)$ .
- Give the standard error of the estimate for  $N = 1000$ .
- Give the number of shots needed for a standard error of 0.01.

## D2-19 · A FULL-LENGTH QUESTION: ONE EXPERIMENT END TO END

A qubit is prepared in  $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$  with  $\theta = \frac{\pi}{2}$  and  $\varphi = \frac{\pi}{3}$ .

- Give the Bloch vector.
- Give the outcome probabilities of an  $X$  measurement.
- Give  $\langle X \rangle$ .
- Give the number of shots needed to estimate  $\langle X \rangle$  to a standard error of 0.02.

## D2-20 · A FULL-LENGTH QUESTION: ONE EXPERIMENT END TO END

A qubit starting in  $|0\rangle$  is driven on resonance by  $H = \frac{\Omega}{2}X$  with  $\Omega = 2$ , held for a time  $t$ , and then measured in the computational basis.

- Give  $|\psi(t)\rangle$ .
- Give  $P(1)$  and  $\langle Z \rangle$  as functions of  $t$ .
- Give the first time at which  $P(1) = \frac{1}{2}$ .
- At that time, give the number of shots needed to estimate  $\langle Z \rangle$  to a standard error of 0.05.

## MODULE

## 3

# Mixed States and Entanglement

20 questions on mixed states and entanglement. Write your reasoning in the space under each one.

## D3-01 · BUILDING A DENSITY OPERATOR, AND TESTING ONE

A source emits  $|0\rangle$  with probability  $\frac{1}{4}$  and  $|+\rangle$  with probability  $\frac{3}{4}$ , and nothing records which.

- Write the density operator as a matrix in the computational basis.
- Is the state pure? Say how you can tell without diagonalising.
- Give the purity, and check it a second way.

## D3-02 · BUILDING A DENSITY OPERATOR, AND TESTING ONE

Three candidate matrices:  $M_1 = \begin{bmatrix} 0.6 & 0.4 \\ 0.4 & 0.4 \end{bmatrix}$ ,  $M_2 = \begin{bmatrix} 0.5 & 0.6i \\ -0.6i & 0.5 \end{bmatrix}$ ,  $M_3 = \begin{bmatrix} 0.5 & 0.3 \\ 0.3 & 0.6 \end{bmatrix}$ .

- Say which of the three are density operators, and which condition each failure breaks.
- For the one that is a state, give its eigenvalues.
- Give its purity, and check it against the eigenvalues.

## D3-03 · BUILDING A DENSITY OPERATOR, AND TESTING ONE

Device A emits  $|+\rangle$  or  $|-\rangle$ , each with probability  $\frac{1}{2}$ . Device B emits  $|0\rangle$  or  $|1\rangle$ , each with probability  $\frac{1}{2}$ .

- Write both density operators.
- Is there any measurement, or any number of copies, that tells the two devices apart?
- Device A is now retuned to emit  $|+\rangle$  with probability  $\frac{3}{4}$ . Give the new matrix, and name a measurement that now separates it from device B.

## D3-04 · PREDICTIONS FROM A DENSITY OPERATOR

A qubit is in the state  $\rho = \begin{bmatrix} 0.7 & 0.2i \\ -0.2i & 0.3 \end{bmatrix}$ .

- Give the three Pauli expectation values.
- Give the outcome probabilities of a  $Z$  measurement.
- Give the purity, and say whether the state is pure.

## D3-05 · PREDICTIONS FROM A DENSITY OPERATOR

A qubit is in the mixture  $\rho = \frac{3}{4}|0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1|$ , and is measured along the direction  $\mathbf{n} = (\sin \alpha, 0, \cos \alpha)$ .

- Give the Bloch vector of the state.
- Give  $p(+1)$  as a function of  $\alpha$ , and evaluate it at  $\alpha = 60^\circ$ .
- At which  $\alpha$  is the measurement a fair coin, and why does the answer not depend on how mixed the state is?

## D3-06 · PREDICTIONS FROM A DENSITY OPERATOR

The observable  $A = X + Z$  is measured on the state  $\rho = \begin{bmatrix} 0.625 & 0.375 \\ 0.375 & 0.375 \end{bmatrix}$ .

- Give  $\langle A \rangle$ .
- Give the variance of  $A$  in this state.
- The mean comes out as 1. Is 1 a reading the instrument can return?

## D3-07 · CHANNELS, DAMPING AND DEPHASING

A qubit prepared in  $|+\rangle$  passes through an amplitude-damping channel of strength  $\gamma$ .

- Give the density matrix after the channel for  $\gamma = 0.36$ .
- Give its purity.
- As  $\gamma$  runs from 0 to 1, what is the smallest purity the output ever has, and at which  $\gamma$ ?

## D3-08 · CHANNELS, DAMPING AND DEPHASING

A qubit prepared in  $|+\rangle$  passes through the phase-flip channel  $\mathcal{E}_Z(\rho) = (1 - p)\rho + pZ\rho Z$ .

- Give the density matrix and the purity for  $p = 0.25$ .
- At which  $p$  is the output the maximally mixed state?
- Give the output at  $p = 1$ , and explain why a channel at full strength has destroyed nothing.

## D3-09 · CHANNELS, DAMPING AND DEPHASING

A device has a relaxation time  $T_1 = 80 \mu\text{s}$  and a pure-dephasing time  $T_\phi = 40 \mu\text{s}$ .

- Give  $T_2$ .
- A qubit is prepared with an excited population and a coherence, and idles for  $20 \mu\text{s}$ . Give the surviving fraction of each.
- What is the largest  $T_2$  this  $T_1$  allows, and what would have to be true of the device to reach it?

## D3-10 · REDUCED STATES AND THE PARTIAL TRACE

Two qubits are in the pure state  $|\psi\rangle = \frac{1}{2}(\sqrt{3}|00\rangle + |11\rangle)$ .

- Give the reduced state of each qubit.
- Give the purity of each.
- Is the pair entangled? Give the reason in one sentence.

## D3-11 · REDUCED STATES AND THE PARTIAL TRACE

Two qubits are in the state  $|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$ .

- Give the reduced state of the first qubit.
- Decide whether the state is entangled, using the amplitude test.
- Give the Schmidt coefficients and the entanglement entropy in bits.

## D3-12 · REDUCED STATES AND THE PARTIAL TRACE

Compare  $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$  with the mixed state  $\rho_{\text{mix}} = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|$ .

- Give the reduced state of the first qubit in each case.
- Give the purity of each joint state.
- Name one measurement whose statistics differ, and give both answers.

## D3-13 · SEPARABILITY, SCHMIDT RANK AND ENTROPY

Three two-qubit pure states: (i)  $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ , (ii)  $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ , (iii)  $\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$ .

- Apply the amplitude test to each.
- Factor the ones that factor.
- For each, say what the reduced state of the first qubit is without computing a partial trace.

## D3-14 · SEPARABILITY, SCHMIDT RANK AND ENTROPY

Two qubits are in the pure state  $|\psi\rangle = \frac{1}{\sqrt{6}}(|00\rangle + |01\rangle + 2|10\rangle)$ .

- Write the coefficient matrix and give  $\rho_A$ .
- Give the two Schmidt coefficients.
- Give the entanglement entropy in bits.

## D3-15 · SEPARABILITY, SCHMIDT RANK AND ENTROPY

A family of two-qubit states,  $|\psi(\theta)\rangle = \cos \theta |00\rangle + \sin \theta |11\rangle$ , with  $\theta$  between 0 and  $90^\circ$ .

- Give the Schmidt coefficients as functions of  $\theta$ .
- At which  $\theta$  is the pair maximally entangled, and at which is it a product?
- Give the entropy at  $\theta = 30^\circ$ .

## D3-16 • BELL CORRELATIONS AND THE CHSH NUMBER

Two qubits are in the singlet  $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ .

- Give  $\langle X \otimes X \rangle$ ,  $\langle Y \otimes Y \rangle$  and  $\langle Z \otimes Z \rangle$ .
- Give the reduced state of each qubit.
- What do the three answers in (a) say about measuring both qubits along the *same* arbitrary direction?

## D3-17 • BELL CORRELATIONS AND THE CHSH NUMBER

Both parties measure  $|\Phi^+\rangle$  along directions in the  $z$ - $x$  plane, at angles from  $z$  of  $A_0 = 0^\circ$ ,  $A_1 = 90^\circ$ ,  $B_0 = 30^\circ$  and  $B_1 = -30^\circ$ . For this state the correlation of two such readings is the cosine of the angle between the directions.

- Give the four correlations.
- Give  $S = \langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle$ , and say whether it violates the classical bound.
- Keeping  $A_0$  and  $A_1$  fixed and the two  $B$  angles symmetric at  $\pm\varphi$ , find the  $\varphi$  that makes  $S$  largest, and the value there.

## D3-18 • BELL CORRELATIONS AND THE CHSH NUMBER

Two parties each hold one qubit of  $|\Phi^+\rangle$ . The second party measures their qubit in the  $X$  basis.

- Rewrite  $|\Phi^+\rangle$  with both qubits in the  $X$  basis.
- Given that the second party obtained  $+1$  and told the first party so, what is the first party's state?
- If the second party says nothing, what is the first party's state — and does the answer depend on whether the second party measured  $X$ , measured  $Z$ , or did nothing at all?

## D3-19 · A FULL-LENGTH QUESTION: ONE PAIR, END TO END

A Bell pair  $|\Phi^+\rangle$  is made. One of the two qubits then passes through a phase-flip channel of strength  $p$  before both are measured along directions in the  $z$ - $x$  plane. The channel multiplies  $\langle X \otimes X \rangle$  and  $\langle Y \otimes Y \rangle$  by  $\eta = 1 - 2p$  and leaves  $\langle Z \otimes Z \rangle$  alone.

- a) With  $A_0 = 0^\circ$ ,  $A_1 = 90^\circ$ ,  $B_0 = 45^\circ$  and  $B_1 = -45^\circ$ , write  $S$  as a function of  $\eta$ .
- b) Evaluate  $S$  at  $p = 0.1$ .
- c) Find the largest  $p$  for which the pair still violates the classical bound.
- d) The state is a mixture of two Bell states and stays entangled for every  $p < \frac{1}{2}$ . What does the gap between that and your answer to (c) say?

## D3-20 · A FULL-LENGTH QUESTION: ONE PAIR, END TO END

A qubit is prepared in  $|+\rangle$  on a device with  $T_1 = 50 \mu\text{s}$  and  $T_2 = 30 \mu\text{s}$ . It is left idle for a time  $t$  and then measured in the  $X$  basis.

- a) Give  $p(+)$  as a function of  $t$ .
- b) Evaluate it at  $t = 20 \mu\text{s}$ .
- c) How many shots are needed to estimate that probability to  $\pm 0.01$  at two standard errors?
- d) Check that the two quoted times are consistent with each other, and say what would have to be measured to know  $T_\phi$ .

## MODULE

## 4

# The Bloch Sphere and Quantum Gates

20 questions on the bloch sphere and quantum gates. Write your reasoning in the space under each one.

## D4-01 · BETWEEN A STATE, A MATRIX AND A POINT

A qubit is prepared in  $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}e^{i2\pi/3}|1\rangle$ .

- a) Give the two angles  $\theta$  and  $\varphi$ .
- b) Give the Bloch vector.
- c) Give the probability of reading 0 in the computational basis, two different ways.

## D4-02 · BETWEEN A STATE, A MATRIX AND A POINT

Tomography on a qubit returns  $\mathbf{r} = (0.6, 0, 0.8)$ .

- a) Is the state pure? Say how you can tell in one line.
- b) Give  $\theta$  and  $\varphi$ , and write the state.
- c) Give the probability of the  $+1$  outcome of an  $X$  measurement and of a  $Z$  measurement.

## D4-03 · BETWEEN A STATE, A MATRIX AND A POINT

Three pure states:  $|a\rangle$  at  $(\theta, \varphi) = (60^\circ, 0)$ ,  $|b\rangle$  at  $(120^\circ, 180^\circ)$ , and  $|c\rangle$  at  $(90^\circ, 0)$ .

- a) Give the three Bloch vectors.
- b) Which pair is orthogonal, and how do you know from the picture alone?
- c) Give  $|\langle c|a\rangle|^2$ .

## D4-04 · A GATE AS A ROTATION

The gate  $R_y(70^\circ)$  is applied to  $|0\rangle$ .

- Write the matrix of  $R_y(\alpha)$ .
- Give the state and the Bloch vector afterwards.
- Give  $p(0)$ , two ways.

## D4-05 · A GATE AS A ROTATION

Consider the gate  $G = \frac{X + Y}{\sqrt{2}}$ .

- Show that  $G$  is a rotation, and give its axis and its angle.
- Give  $G|0\rangle$ .
- Give  $G|+i\rangle$ , using the geometry rather than the matrices.

## D4-06 · A GATE AS A ROTATION

A qubit in  $|+\rangle$  has the gate  $T$  applied to it.

- Give the Bloch vector afterwards.
- Give the probability of reading 0 in the computational basis.
- A Hadamard is now applied as well. Give the probability of reading 0, and say what changed.

## D4-07 · COMPOSING A SEQUENCE OF GATES

The gates  $H$ , then  $S$ , then  $H$  are applied to  $|0\rangle$ , in that order.

- Write the matrix product that represents the circuit.
- Follow the Bloch vector through the three gates.
- Name the state at the end, and say what single rotation the whole circuit is.

## D4-08 • COMPOSING A SEQUENCE OF GATES

Two gates,  $H$  and  $S$ , and one input  $|0\rangle$ .

- Give the state when  $H$  runs first, and when  $S$  runs first.
- How far apart are the two results on the sphere, and what is the overlap?
- Say in one sentence why the two orders disagree.

## D4-09 • COMPOSING A SEQUENCE OF GATES

The four gates  $X$ ,  $Z$ ,  $X$ ,  $Z$  are applied in that order.

- Give the net unitary of the sequence.
- What does the sequence do to any Bloch vector?
- A control qubit is added, so that the whole four-gate sequence runs only when the control is  $|1\rangle$ . The control starts in  $|+\rangle$ . What is the control at the end?

## D4-10 • DECOMPOSING AND SYNTHESISING A GATE

The  $S$  gate,  $S = \text{diag}(1, i)$ .

- Write  $S$  as a phase times a rotation about  $\hat{z}$ .
- Give its Euler form  $e^{i\alpha} R_z(\phi) R_y(\theta) R_z(\lambda)$ .
- Is the controlled- $S$  the same circuit as the controlled- $R_z(\pi/2)$ ? If not, what is the difference?

## D4-11 • DECOMPOSING AND SYNTHESISING A GATE

A gate  $U(\theta, \phi, \lambda)$  is wanted that prepares the state at  $(\theta, \varphi) = (60^\circ, 45^\circ)$  from  $|0\rangle$ .

- Give the parameters  $\theta$  and  $\phi$ .
- Is  $\lambda$  determined by the requirement? Say why.
- With  $\lambda = 0$ , write the four entries of the matrix.

## D4-12 · DECOMPOSING AND SYNTHESISING A GATE

A machine offers only  $H$ ,  $S$ ,  $T$  and CNOT.

- How many  $T$  gates make a  $Z$ ? Show the chain.
- Is  $\{H, S, \text{CNOT}\}$  on its own universal? Say why or why not.
- A one-qubit gate is to be approximated to  $\varepsilon$ , and the gate count grows like  $\log^2(1/\varepsilon)$ . By what factor does the count grow when  $\varepsilon$  drops from  $10^{-2}$  to  $10^{-4}$ ?

## D4-13 · TWO-QUBIT GATES, AND WHICH QUBIT IS WHICH

The two-qubit state is  $|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$ , written  $|q_1q_0\rangle$ .

- Give  $(I \otimes X)|\psi\rangle$ .
- Give  $(X \otimes I)|\psi\rangle$ .
- Name one measurement outcome whose probability tells the two apart.

## D4-14 · TWO-QUBIT GATES, AND WHICH QUBIT IS WHICH

The state  $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$  is handed to a CNOT.

- Give the result when the control is  $q_0$  and the target is  $q_1$ .
- Give the result when the control is  $q_1$  and the target is  $q_0$ .
- Which of the two outputs is entangled? Give its entanglement entropy in bits.

## D4-15 · TWO-QUBIT GATES, AND WHICH QUBIT IS WHICH

The controlled- $Z$  gate,  $\text{CZ} = \text{diag}(1, 1, 1, -1)$ , is applied to  $|+\rangle \otimes |+\rangle$ .

- Give the output state.
- Is it entangled? Give the reduced state of one qubit and the entropy.
- CZ moves no probability at all. Reconcile that with the answer to (b).

## D4-16 · TWO-QUBIT GATES, AND WHICH QUBIT IS WHICH

A chip couples  $q_0$  to  $q_1$  and  $q_1$  to  $q_2$ , and nothing else. A CNOT is wanted with  $q_0$  as control and  $q_2$  as target.

- Give the SWAP gate as a matrix, in the ordering  $|q_1 q_0\rangle$ .
- How many CNOTs does the whole job cost if the qubits may be left in their new positions?
- And if they must be returned to where they started?
- A reversible half adder must also write  $s = a \oplus b$  and  $c = ab$ . Give an embedding that preserves its inputs, and name the gates that write the two outputs.

## D4-17 · ENTANGLING POWER AND UNIVERSALITY

From  $|00\rangle$ , the gate  $R_y(\theta)$  is applied to  $q_0$  and then a CNOT with  $q_0$  as control and  $q_1$  as target.

- Give the output state in terms of  $\theta$ .
- Give the entanglement entropy at  $\theta = 120^\circ$ .
- At which  $\theta$  is the output a product state?
- A further one-qubit gate is applied to each qubit. What happens to the entropy?

## D4-18 · ENTANGLING POWER AND UNIVERSALITY

Four proposed gate sets: (i) every one-qubit gate; (ii)  $\{H, S, \text{CNOT}\}$ ; (iii)  $\{H, S, T, \text{CNOT}\}$ ; (iv) every one-qubit gate together with CNOT.

- Which of the four can produce an entangled state at all?
- Which are universal, and in which sense — exactly or to any accuracy?
- Give one sentence saying what "universal" does not promise.

## D4-19 · A FULL-LENGTH QUESTION: ONE CIRCUIT, END TO END

A single qubit starts in  $|0\rangle$ . The circuit applies  $H$ , then  $T$ , then  $H$ , and the qubit is measured in the computational basis.

- a) Give the Bloch vector after each of the three gates.
- b) Give  $p(0)$  exactly, and as a decimal.
- c) How many shots are needed to estimate that probability to  $\pm 0.01$  at two standard errors?
- d) The  $T$  is replaced by  $T^\dagger$ . What happens to  $p(0)$ , and why?

## D4-20 · A FULL-LENGTH QUESTION: ONE CIRCUIT, END TO END

Two qubits start in  $|00\rangle$ , written  $|q_1 q_0\rangle$ . The circuit applies  $H$  to  $q_0$ , then a CNOT with  $q_0$  as control and  $q_1$  as target, then  $Z$  to  $q_1$ .

- a) Give the state after each of the three gates, and name the final one.
- b) Give the two reduced states, and the entanglement in bits.
- c) Give the probabilities of the four computational-basis outcomes.
- d) A colleague claims the circuit produced  $|\Phi^+\rangle$ . Name one measurement setting that decides the question, and say how many shots it needs in the ideal case.

## MODULE

## 5

# Circuits and Protocols

20 questions on circuits and protocols. Write your reasoning in the space under each one.

## D5-01 · READING AND COUNTING A CIRCUIT

A GHZ state on eight qubits is prepared two ways. The **chain** applies  $H$  to  $q_0$  and then  $\text{CNOT}_{0 \rightarrow 1}$ ,  $\text{CNOT}_{1 \rightarrow 2}$ , and so on up to  $\text{CNOT}_{6 \rightarrow 7}$ . The **tree** applies  $H$  to  $q_0$ , then  $\text{CNOT}_{0 \rightarrow 1}$ , then  $\text{CNOT}_{0 \rightarrow 2}$  and  $\text{CNOT}_{1 \rightarrow 3}$  together, then four more together.

- Give the total gate count and the two-qubit count for each.
- Give the depth of each.
- A two-qubit gate takes 400 ns and a one-qubit gate 40 ns. Give the duration of each.

## D5-02 · READING AND COUNTING A CIRCUIT

A three-qubit circuit maps  $q_k$  to classical bit  $c_k$ , and one run prints the string 011.

- Give the values read on each wire, and name the wire drawn at the top.
- Give the ket and the state-vector index this string names.
- Write the matrix of  $\text{CNOT}_{0 \rightarrow 1}$  on two qubits in the ordered basis  $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ .

## D5-03 · READING AND COUNTING A CIRCUIT

From  $|00\rangle$ , a circuit applies  $H$  to  $q_0$ , then  $\text{CNOT}_{0 \rightarrow 1}$ , then  $Z$  to  $q_1$ .

- Give the state after each of the three gates.
- Name the final state.
- Give the four measurement probabilities, and the depth of the circuit.

## D5-04 · SHOTS, ERROR BARS AND SIMULATION COST

A circuit produces an outcome with true probability  $p = 0.2$ .

- Give the standard error of the estimated probability after 1000 shots.
- Give the shots needed for a standard error of 0.002.
- The measured value comes out at 0.31 after 40,000 shots. Say what that does and does not mean.

## D5-05 · SHOTS, ERROR BARS AND SIMULATION COST

An exact simulation stores each complex amplitude in 16 bytes.

- Give the memory a 34-qubit state vector needs.
- Give the largest  $n$  whose state vector fits in  $1 \text{ TB} = 10^{12}$  bytes.
- Give the memory the circuit matrix of a 15-qubit circuit needs, and say why circuits are checked on states instead.

## D5-06 · SHOTS, ERROR BARS AND SIMULATION COST

A one-qubit dynamic circuit applies  $H$  to a qubit in  $|0\rangle$ , measures it into  $c_0$ , applies  $X$  if  $c_0 = 1$ , and measures again into  $c_1$ .

- Give the joint distribution of  $(c_1, c_0)$ .
- Give it again with the conditional  $X$  removed.
- Give it again with the conditional  $X$  kept and one further  $H$  inserted before the second measurement.

## D5-07 · REWRITING AND ROUTING FOR A MACHINE

A circuit has 12 CNOTs and 20 one-qubit gates. The machine offers only  $R_z(\lambda)$ ,  $R_x(\pi/2)$  and CZ. Each CNOT becomes  $H CZ H$ , each  $H$  becomes three rotations, and each of the original one-qubit gates becomes three rotations.

- Give the instruction count after translation, and the two-qubit count.
- A two-qubit gate fails with probability 0.008 and a one-qubit gate with 0.0003. Give the probability that no gate fails.
- Say which of the two counts in part (a) the answer to part (b) depended on.

## D5-08 · REWRITING AND ROUTING FOR A MACHINE

A chip has five qubits joined in a line,  $Q_0 - Q_1 - Q_2 - Q_3 - Q_4$ , and the circuit asks for a CNOT between  $Q_0$  and  $Q_4$ .

- Give the number of two-qubit gates the compiler needs, leaving the qubits permuted at the end.
- Give it again if the original layout has to be restored.
- Give it again if the chip is a ring, with  $Q_4$  also joined to  $Q_0$ .

## D5-09 · REWRITING AND ROUTING FOR A MACHINE

A compiler returns two valid versions of one circuit. Version A has 40 two-qubit gates and depth 30; version B has 52 and depth 18. Every layer contains at least one two-qubit gate, a two-qubit gate takes 400 ns and fails with probability 0.01, and the qubits have  $T_2 = 60 \mu s$ .

- Give the duration of each version.
- Give the probability that no gate fails, for each.
- Combine the two effects with a factor  $e^{-T/T_2}$  and say which version to run.

## D5-10 · TELEPORTATION: BRANCHES, CORRECTIONS AND RESOURCES

Alice teleports  $|\psi\rangle = 0.6|0\rangle + 0.8|1\rangle$  and reads  $m_1 = 1, m_0 = 0$ .

- Give the state Bob holds before any correction.
- Give the correction he applies, and verify that it returns  $|\psi\rangle$ .
- Give the fidelity Bob would have if he applied nothing.

## D5-11 · TELEPORTATION: BRANCHES, CORRECTIONS AND RESOURCES

Alice teleports an arbitrary qubit  $|\psi\rangle$  with Bloch vector  $\mathbf{r}$ .

- Give the probability of each of the four branches.
- Give the four Bloch vectors Bob's qubit could carry, and their average.
- Say whether Bob can tell that Alice ran the protocol at all, before the bits arrive.

## D5-12 · TELEPORTATION: BRANCHES, CORRECTIONS AND RESOURCES

A three-qubit state is to be teleported, and a colleague proposes saving bandwidth by sending only  $m_1$  for each qubit.

- Give the entangled pairs and classical bits the honest protocol needs.
- With only  $m_1$  sent, give the state Bob ends up with, as a channel applied to  $\rho$ .
- Give the resulting average fidelity over inputs spread across the sphere, and say what it shows.

## D5-13 · TELEPORTATION: BRANCHES, CORRECTIONS AND RESOURCES

A group reports an average teleportation fidelity of  $F_{\text{avg}} = 0.78$  over inputs spread across the sphere, using pairs of singlet fraction  $f$  and the relation  $F_{\text{avg}} = (2f + 1)/3$ .

- Say whether the result beats the classical benchmark.
- Give the singlet fraction the result implies.
- Give the singlet fraction that would be needed for  $F_{\text{avg}} = 0.9$ , and name two things the report must state besides the fidelity.

**D5-14 · GROVER: ANGLE, ITERATIONS AND PROBABILITY**

Four qubits carry  $N = 16$  candidates, and exactly one is marked.

- Give the angle  $\theta$  in degrees.
- Give the exact optimum  $r_*$  and the whole number nearest it.
- Give the success probability at that whole number and at the one below it.

**D5-15 · GROVER: ANGLE, ITERATIONS AND PROBABILITY**

Six qubits carry  $N = 64$  candidates.

- With  $M = 4$  marked, give  $\theta$ , the best whole  $r$ , and  $P$  at it.
- With  $M = 1$  marked, give the same three numbers.
- Give the ratio of the two iteration counts, and say what it shows.

**D5-16 · GROVER: ANGLE, ITERATIONS AND PROBABILITY**

Ten qubits carry  $N = 1024$  candidates, one of them marked. A run is scheduled for as many iterations as the coherence time allows.

- Give the best whole number of iterations and the success probability there.
- Give the success probability after 50 iterations.
- Give it after 75, and say what the three answers together mean for the schedule.

**D5-17 · GROVER: ANGLE, ITERATIONS AND PROBABILITY**

Two qubits carry  $N = 4$  candidates and the marked one is  $x = 2$ , that is  $|10\rangle$ . The register starts in the uniform superposition and the oracle target is prepared in  $|-\rangle$ .

- Give the state of the register after one oracle call.
- Give the probability of measuring the marked state at that point.
- Give the best whole number of iterations and the success probability there.

## D5-18 · ASSESSING A RESOURCE CLAIM

A report states: "our quantum algorithm searches a ten-million-row customer database quadratically faster than any classical method".

- a) Name the five components a resource claim must state, and say which the sentence states.
- b) Say what the input model has to be for the quadratic saving to hold, and what that costs here.
- c) Say what the correct classical baseline for this task is.

## D5-19 · ASSESSING A RESOURCE CLAIM

One quantum oracle call takes  $20\ \mu\text{s}$  on the machine available; the same predicate takes  $5\ \text{ns}$  to evaluate on a classical processor. Grover needs about  $\frac{\pi}{4}\sqrt{N}$  calls and the classical search about  $N/2$ .

- a) Give both total times at  $N = 10^6$ .
- b) Give the problem size at which the two are equal.
- c) Say what would move that size, in each direction.

## D5-20 · A FULL-LENGTH QUESTION: ONE PROTOCOL, END TO END

A group reports finding a marked item among  $N = 4096$  candidates on hardware. Their oracle compiles to 65 two-qubit gates and their diffusion operator to 62. Two-qubit gates fail with probability 0.01 and there is no error correction.

- a) Give  $\theta$ , the exact optimum  $r_*$ , and the best whole number of iterations.
- b) Give the success probability at that count, in the ideal circuit model.
- c) Give the total number of two-qubit gates the run needs, and the probability that none of them fails.
- d) Say how many shots would be needed to distinguish an ideal run from a uniformly random one, and what the answer to part (c) does to that plan.
- e) Say which of the five components of a resource claim this report has and has not established.

## MODULE

## 6

# Quantum Algorithms

20 questions on quantum algorithms. Write your reasoning in the space under each one.

## D6-01 · KICKBACK AND INTERFERENCE

The Deutsch circuit is run with the constant oracle  $f(x) = 1$ . The query qubit starts in  $|0\rangle$  and the target in  $|1\rangle$ , both are sent through a Hadamard, the oracle acts, and a Hadamard is applied to the query qubit alone.

- Give the state of the two qubits just after the oracle.
- Give the state of the query qubit after the last Hadamard.
- Give the reading and say what it proves and what it does not.

## D6-02 · KICKBACK AND INTERFERENCE

The Deutsch–Jozsa circuit is run on  $n = 3$  query qubits with the oracle  $f(x_2x_1x_0) = x_0$ .

- Say whether the oracle satisfies the promise, and why.
- Give the amplitude of  $|000\rangle$  at the output, and the state the circuit actually produces.
- Give the number of queries an exact classical algorithm needs in the worst case.

## D6-03 · KICKBACK AND INTERFERENCE

A single control qubit in  $|+\rangle$  controls the gate  $S = \text{diag}(1, i)$ , whose target is prepared in  $|1\rangle$ .

- Give the eigenphase  $\varphi$  that the control picks up.
- Give the state of the control qubit after the gate, and the probability of reading 0 after a Hadamard.
- Say why one control qubit cannot tell this phase from  $\varphi = 3/4$ , and what measurement would.

## D6-04 · PROMISE PROBLEMS AND QUERY COUNTS

A promised function on  $n = 8$  bits is either constant or balanced.

- Give the exact classical worst-case query count and the quantum query count.
- Give the number of distinct classical queries needed to be wrong with probability below  $10^{-3}$ .
- Say what happens to each of those three counts as  $n$  grows.

## D6-05 · PROMISE PROBLEMS AND QUERY COUNTS

The Deutsch–Jozsa circuit is run on  $n = 4$  query qubits with a function that returns 1 on exactly six of its sixteen inputs. The promise has been broken.

- Give the amplitude of  $|0000\rangle$  at the output.
- Give the probability of reading 0000.
- Say what a reading of 0000 and a reading of anything else each allow you to conclude.

## D6-06 · PROMISE PROBLEMS AND QUERY COUNTS

On three query qubits the oracle computes  $f(x_2x_1x_0) = x_0 \oplus x_2$  into the target, using two CNOT gates.

- Say whether the promise holds.
- Give the string the circuit prints, and the probability of printing it.
- Give the depth of the oracle and say whether the two CNOTs can run together.

## D6-07 · THE TRANSFORM: AMPLITUDES AND COST

The three-qubit Fourier transform,  $Q = 8$ , is applied to the basis state  $|6\rangle$ .

- Give the modulus of every output amplitude.
- Give the angle the phase advances by as  $k$  increases by one.
- Give the eight measurement probabilities, and say what a single run reveals about the input.

## D6-08 · THE TRANSFORM: AMPLITUDES AND COST

The exact Fourier transform circuit is to be built on  $n = 6$  qubits.

- a) Give the number of Hadamards, the number of controlled rotations and the number of swaps.
- b) Give the smallest rotation angle in the circuit.
- c) A classical fast Fourier transform on 64 stored numbers costs about  $Q \log_2 Q$  operations. Give that count and say why the two numbers should not be compared.

## D6-09 · THE TRANSFORM: AMPLITUDES AND COST

The state  $\frac{1}{\sqrt{2}}(|0\rangle + |4\rangle)$  on three qubits is sent through the Fourier transform,  $Q = 8$ .

- a) Give the eight output amplitudes.
- b) Give the eight measurement probabilities.
- c) Say what one run tells you about the input, and how many runs a full description of the output would take.

## D6-10 · PHASE ESTIMATION: PRECISION AND COST

Phase estimation is run with  $t = 4$  counting qubits on a unitary whose eigenphase is  $\varphi = 5/16$ .

- a) Give the reading and its probability.
- b) Give the four bits the counting register holds.
- c) Give the number of applications of  $U$  the run needs.

## D6-11 · PHASE ESTIMATION: PRECISION AND COST

Phase estimation is run with  $t = 3$  counting qubits on a unitary whose eigenphase is  $\varphi = 0.1$ .

- a) Give the two readings that bracket the true phase.
- b) Give the probability of each of them.
- c) Compare their total with the guaranteed floor, and give the error of the best estimate.

## D6-12 · PHASE ESTIMATION: PRECISION AND COST

A phase is to be estimated to six correct binary digits, with a failure probability of at most 0.05.

- Give the number of counting qubits.
- Give the number of applications of  $U$  that needs.
- Say what changes if the failure probability is tightened to 0.005, and what changes if a seventh digit is wanted instead.

## D6-13 · PHASE ESTIMATION: PRECISION AND COST

Phase estimation is run with  $t = 12$  counting qubits, on a unitary whose own circuit is 800 gates.

- Give the number of applications of  $U$  and the gate count they come to.
- Give the gate count of the inverse Fourier transform.
- Say what fraction of the circuit the transform is, and what a rewrite that halved it would buy.

## D6-14 · ORDER FINDING AND ITS CLASSICAL STEPS

Take  $N = 33$  and  $a = 5$ .

- Give the order of 5 modulo 33.
- Check the two conditions the factoring step needs.
- Give the two factors, and verify them.

## D6-15 · ORDER FINDING AND ITS CLASSICAL STEPS

Order finding is run on  $N = 21$  with  $a = 2$ . Take the idealised case in which every run samples one value of  $s$  uniformly and the continued fractions return the reduced form of  $s/r$ .

- Give the order, the eigenphases, and the probability of each.
- For each measured  $s$ , give the candidate order the continued fractions return and say whether it passes the check.
- Give the probability that one run succeeds.

## D6-16 · ORDER FINDING AND ITS CLASSICAL STEPS

An order-finding run on  $N = 15$  with  $a = 2$  uses  $t = 8$  counting qubits and returns the reading  $y = 192$ .

- Give the fraction  $y/Q$  and its continued-fraction convergents.
- Give the candidate order and confirm it.
- Give the two factors of fifteen.

## D6-17 · ORDER FINDING AND ITS CLASSICAL STEPS

An order-finding run on  $N = 21$  with  $a = 2$  uses  $t = 9$  counting qubits and returns  $y = 427$ .

- Give the convergents of  $y/Q$ .
- Give the candidate order and confirm it.
- Say why  $t = 9$  was chosen rather than  $t = 5$ .

## D6-18 · ASSESSING AN ALGORITHMIC CLAIM

A summary states: "the Deutsch–Jozsa algorithm gives an exponential speedup over classical computers."

- Name the five components a resource claim has to state, and say which of them this sentence states.
- Rewrite the sentence so that it is true.
- Give the largest honest ratio for  $n = 20$ , and say against which classical algorithm.

## D6-19 · ASSESSING AN ALGORITHMIC CLAIM

A resource estimate for factoring a 1024-bit modulus is to be sketched. Take a reversible modular multiplier to cost about  $L^2$  gates, and a counting register of  $t = 2L$  qubits.

- Give the number of controlled modular multiplications and the gates they come to.
- Give the gate count of the inverse Fourier transform on that register.
- Say which of the two an estimate is really an estimate of, and what the hardware model adds.

## D6-20 · A FULL-LENGTH QUESTION: ONE ALGORITHM, END TO END

Factor  $N = 21$  by order finding, with the base  $a = 2$ . Take the counting register to be as short as the uniqueness condition allows, and assume every reading is exact.

- Give the counting register size and the number of applications of the modular multiplier.
- Give the order, the eigenphases, and the probability that one run yields a usable reading.
- Take the reading  $y = 427$  and recover the order from it.
- Give the two factors, and verify them.
- State the claim this run supports and the claim it does not, using the five components.



## Quantum Computing · Student Workbook

| VERSION | DATE           | CHANGES        |
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